



Introduction to Advanced Process Control (APC)

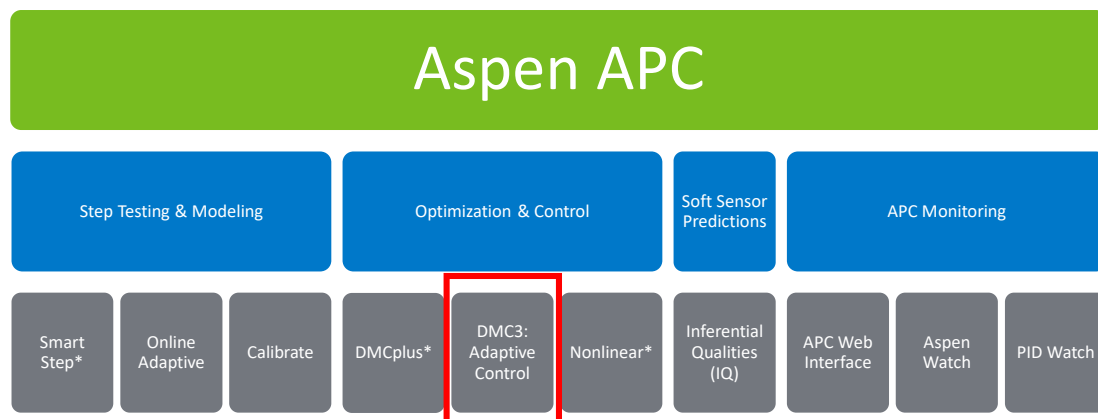


Lesson Objectives

Upon completion of this lesson, you will be able to:

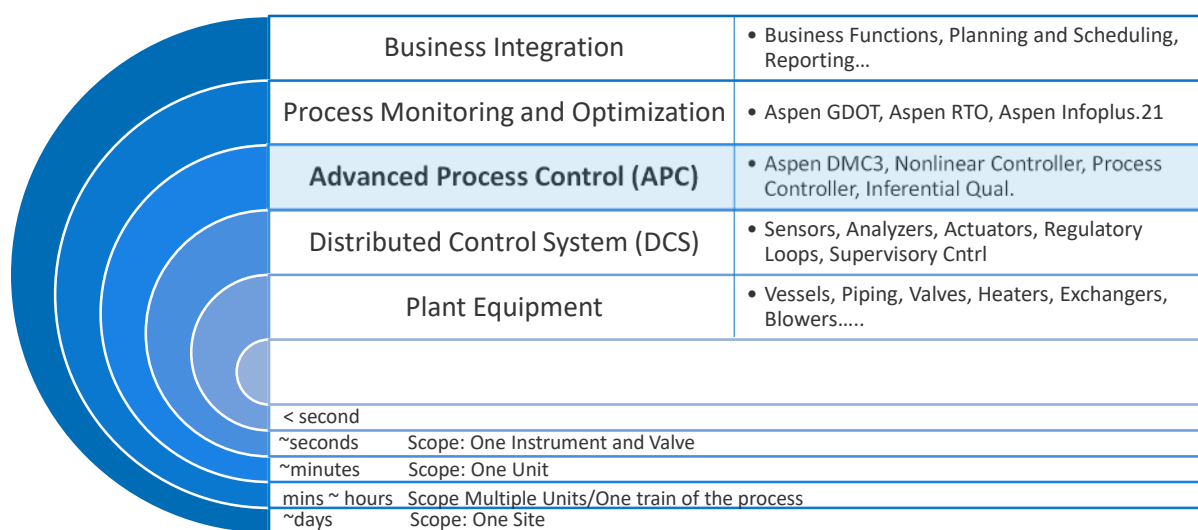
- Learn about Overview of the APC techniques
- Learn about APC application in the petrochemical industry
- Understand Key benefit of APC implementation in petrochemical operations
- Understand overview of the Integration with Aspen GDOT (Generic Dynamic Optimization)

Overview of the APC techniques



*Not supported in current versions

Where is APC in the Control and Optimization Hierarchy?

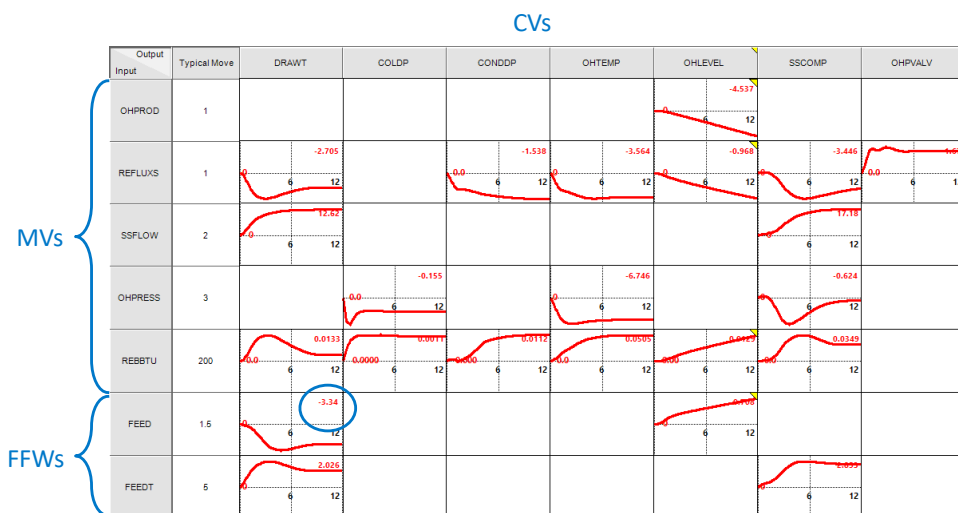


Key benefits of APC implementation in petrochemical operations

- DMC3 updates Steady state Optimum Operating point/Optimum Dynamic Move plan every cycle
 - No distractions – fully focused on the process
- Three computations:
 1. **Where is the process Going?**
Reads the current process values and uses the model to predict what the process conditions will be in the future
 2. **Where is the Optimum operating point?**
Calculates the optimum operating point for the process based on process values and economics
 3. **How can the process be Moved to the Optimum Operating Point?**
Calculates a set of adjustments to the manipulated variables that will determine the optimum path to move the process towards the economic optimum
 - Rejecting disturbances
 - Responding to specification changes by operators
- Can reduce load of the operator

Key benefits of APC implementation in petrochemical operations

- Dynamic Response Curve for each Input-Output Pair
- X-Axis extends to Time to SS
- Y-Axis extends to SS Gain, as written on top right corner of each curve

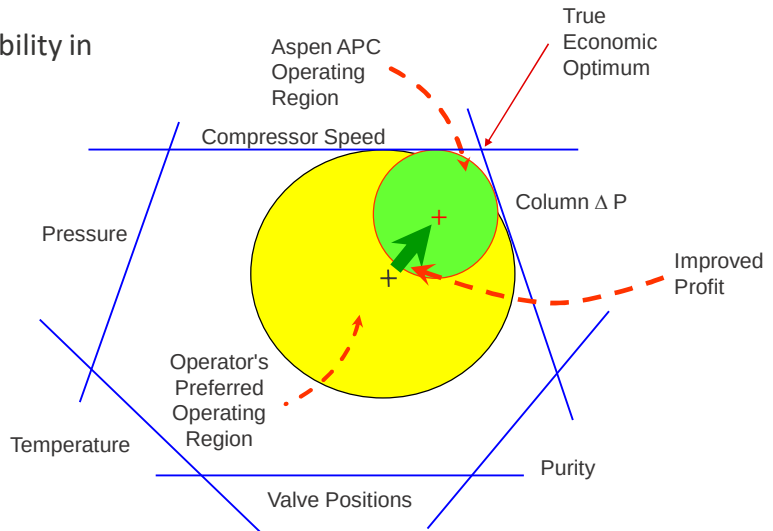


Key benefits of APC implementation in petrochemical operations

3-5% Improvement in Unit Profitability in

5 areas:

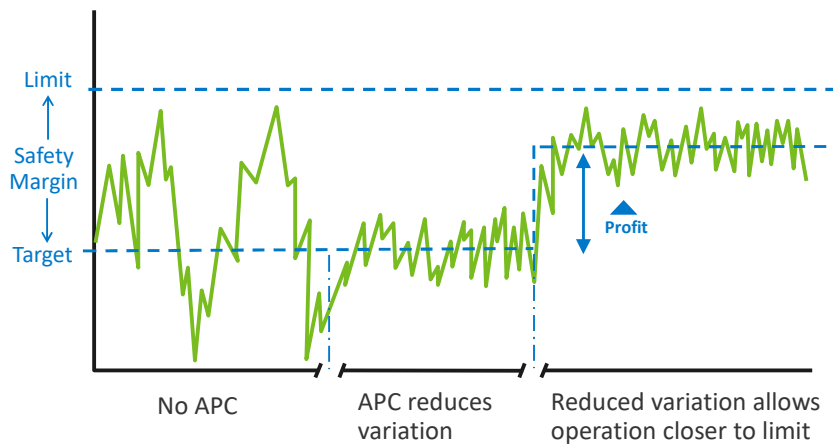
1. Maximize feed and products
2. Minimize utilities
3. Push to the most profitable constraints



Key benefits of APC implementation in petrochemical operations

4. Reduce product quality variation by roughly 50%

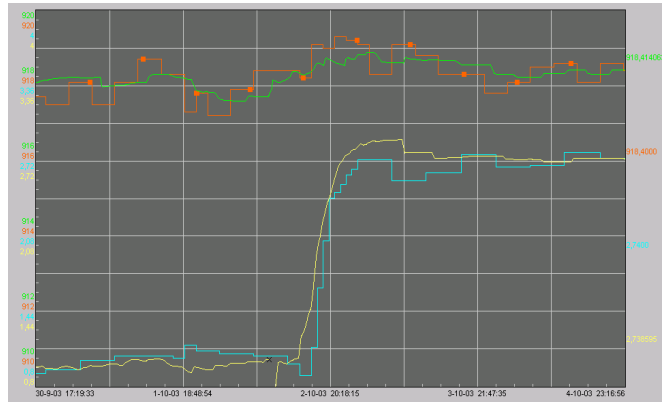
Figure: Product variation has a large range before APC is implemented. The target for this variable must be set so that on its worst excursion, the value does not exceed the upper limit. Once APC is implemented, the variability is dramatically reduced due to its preemptive action through model predictive capability. This allows the target for the variable to be moved closer to the upper limit, which drives increase in profit.



Key benefits of APC implementation in petrochemical operations

5. Respond faster to specification changes or disturbances

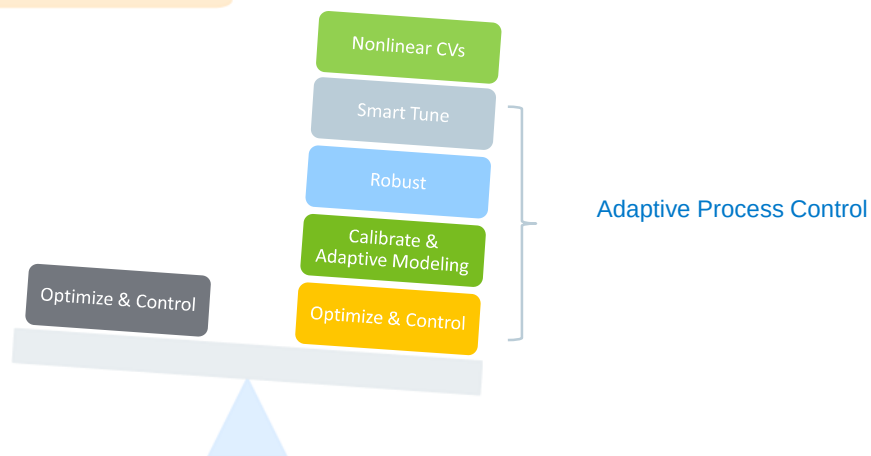
Figure: The blue/yellow lines increasing show that a disturbance has occurred that would affect operating conditions. However, the other variable's response in red/green lines was not so volatile and maintained reasonably. This rapid response by the APC controller minimizes off-spec product, thus improving control and increasing profitability.



Aspen DMC3 vs DMCplus

Aspen DMCplus

Aspen DMC3



What is Aspen Inferential Quality (IQ)?

- Aspen Inferential Qualities (AspenIQ for short) is an AspenTech product that provides all of the tools to implement linear inferential sensors for industrial processes.
- Used as information or feedback to engineers or operators
 - Some batch processes
 - Some event-driven processes
- Used with Advanced Process Control (i.e., with the Aspen DMCplus Controller) to realize even more benefits by using as Measurement of the CV variable.
 - Refining processes
 - Petrochemical processes
 - Chemical processes
 - etc.

Examples of Inferential Sensors

- Refinery petroleum fractionators
 - Petroleum fraction boiling points (IBP, 90%, 95%, EP), Flash Point
 - Petroleum fraction “cold” properties (Freeze point, Cloud point)
- Light ends distillation (refining or petrochemical)
 - Product impurity (C5+, C4-, etc.)
- Emissions Monitoring
 - NOX, CO, particulates

Applications in the petrochemical industry

Wide applicability/Achievements in the petrochemical/Process industry

Refining

Typical Petro Chemical Processes: CDU, Reformer, Hydro Treater, FCC etc..

Utility (Steam/Power plant)

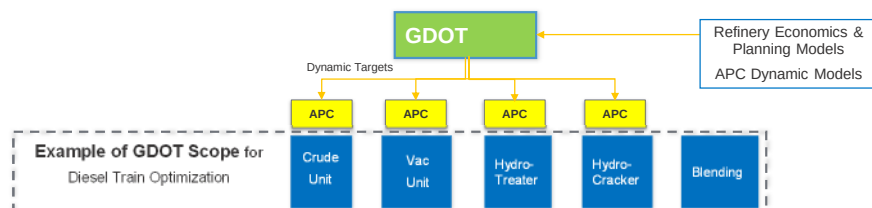
Chemical

Olefin plant, Specialty Chemicals(Continuous process), Aromatics, Polymer plant,

Air separation, etc...

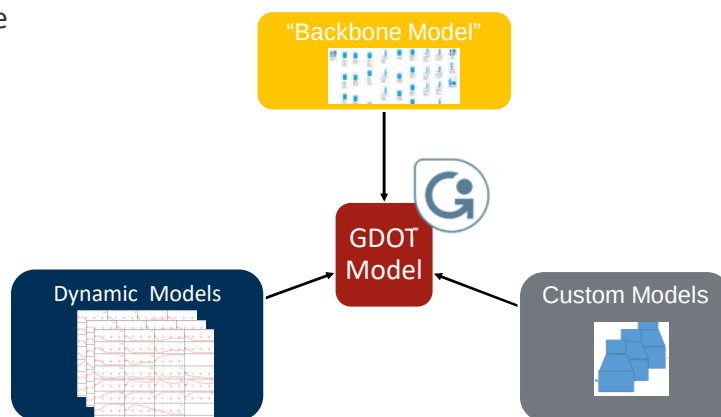
Integration with Aspen GDOT (Generic Dynamic Optimization)

- Combines GDOT template model with planning(PIMS) and dynamic APC models
- Is based on auto-calibrated models using patented dynamic data reconciliation & parameter estimation technology to maximize advantage of minute-by-minute process feedback.
- Runs in a real time environment (typically every minute just like APC controllers, frequency can be adjusted), but the scope of optimization is much larger.



GDOT Combines Planning and APC models

- Using GDOT template, models are generated suitable for online use
 - Fractionators
 - Distillation columns
 - Hydrotreaters
 - Mixers/Splitters
 - Compressors
- GDOT models have additional structure for enabling online adaptation.



Aspen GDOT vs Steady State RTOs



FASTER OPTIMIZATION

- Zero waiting for steady state
- Simplified models in downstream

Cycle Duration:

- Traditional RTO: $60 + 30 = 90$ min
- Aspen GDOT: $0 + 5 = 5$ min



INTEGRATED SOLUTION

- Aligned with Planning by design (top-down and bottom-up)
- Utilizes model dynamics from APC
- Consistent optimization offline & online



EASIER TO MAINTAIN

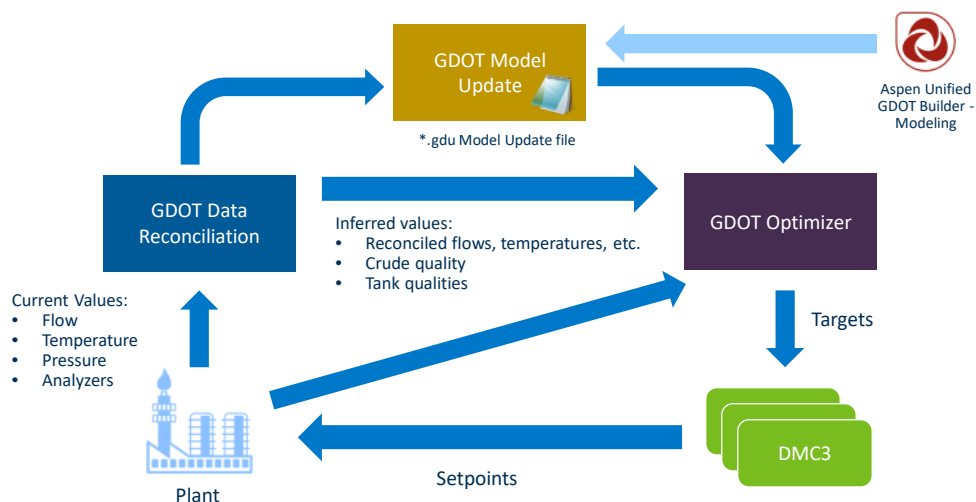
- Less fragile, more autonomous
- Model updates similar to PIMS
- Manageable with APC skillset
- Higher utilization rate



EASIER TO EXPAND

- Parallel olefin trains
- Relevant downstream units
- Integration with GDOT optimizers in upstream Refinery

GDOT Application Structure



Typical Refining GDOT Applications

- Middle distillate (MD) production
 - Coordinate CDU, HDS, Hydrocracking and blending of Diesel and Jet Fuel
- Naphtha processing, aromatics and gasoline blending
 - Coordinate Naphtha fractionation, reforming, Isomerization and blending
- Residue processing
 - Coordinate and optimize carbon rejection and cracker feed treatment
- FCCU and olefins
 - Coordinate FCC, Alkylation, IC3/4 and NC3/4 processing
- Hydrocracking
 - Optimize severity and feed quality subject to unit constraints
- Hydrogen, fuel gas and steam networks
 - Coordinate producers and consumers
- Heat integration
- GDOT models are modular so the above applications can be built as stand-alone optimizers or combined as appropriate for the specific configuration

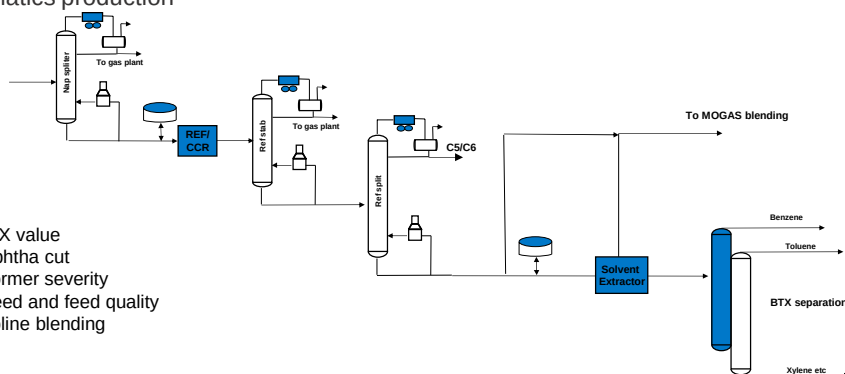
Gasoline/Aromatics Production Systems

Typical Benefits

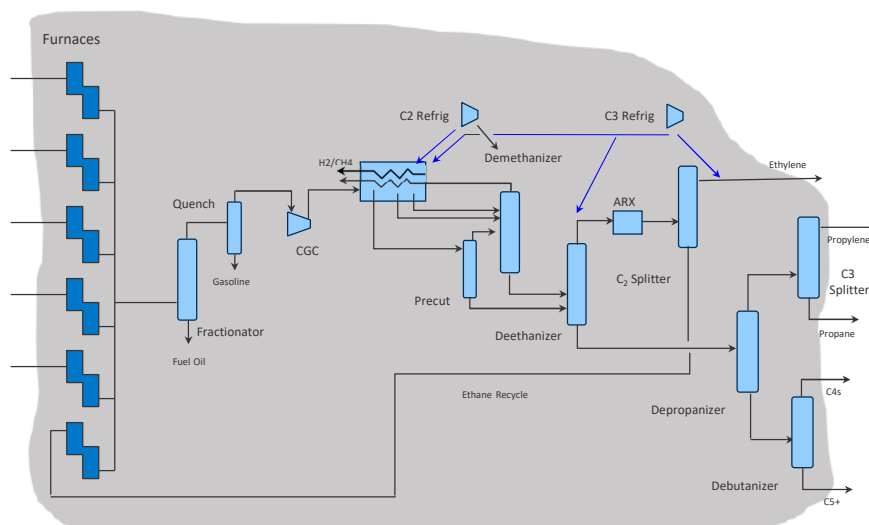
- Reduce quality giveaway
- Optimize benzene production/removal (non-aromatics)
- Optimize parallel units
- Maximize aromatics production

Objectives:

- Maximize BTX value
- Optimize Naptha cut
- Optimize reformer severity
- Coordinate feed and feed quality
- Honour Gasoline blending constraints



Typical GDOT Scope for Single Ethylene train

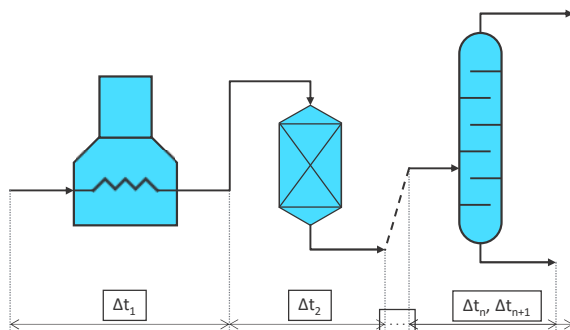


- Feed System
- Furnaces
- Fractionator columns
- Raw gas compressors
- Cold Boxes
- Demethanizers
- Deethanizers
- C2 Splitters
- Depropanizer
- Debutanizer
- C3 Splitters

Appendix

There are Two Kinds of Real Time Optimization...

Large process plants: rarely settled in every section



Steady State Real Time Optimization

- Does not model dynamic response factors (Δt_i)
- Can't optimize when the process is in flux
- Resorts to "steady enough" state detection
 - Waiting time ~1 hr. for Olefins plant
 - Still a compromise b/w accuracy and speed
 - Difficult to cover larger envelopes

Dynamic Real Time Optimization

Aspen GDOT

- Uses and continuously updates dynamic gains, overwriting APC gains for a calculation more consistent with the mass/energy balances.



Model Building from Process Step test data



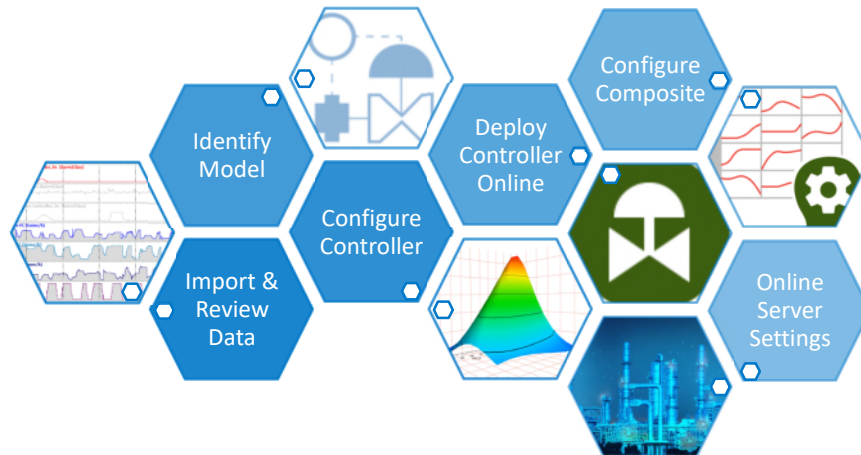
Lesson Objectives

Upon completion of this lesson, you will be able to:

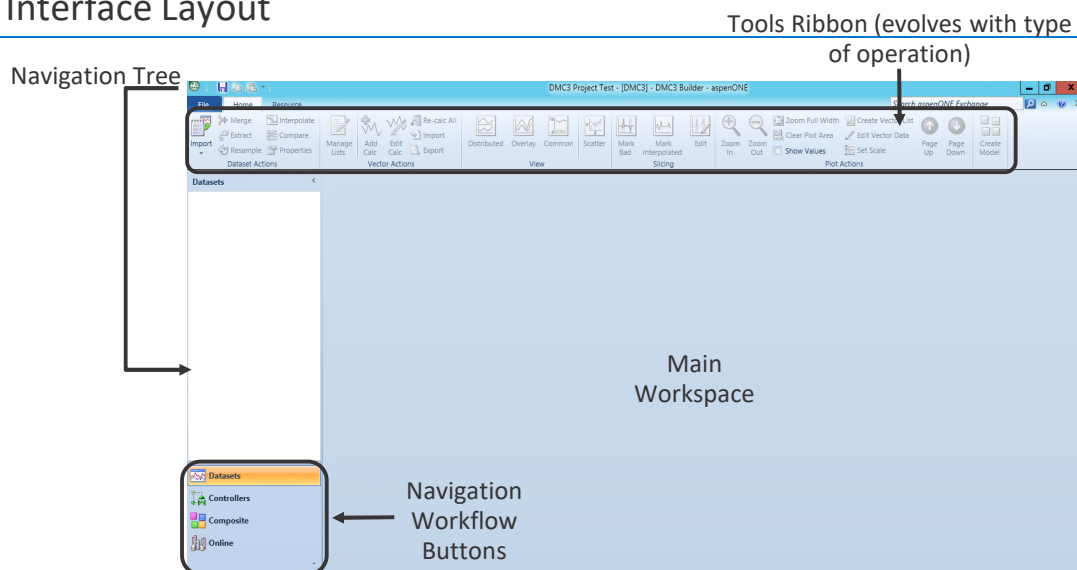
- Model building by using DMC3 Builder (Mini-Workshop)
- Controller Deployment work flow

DMC3 Builder

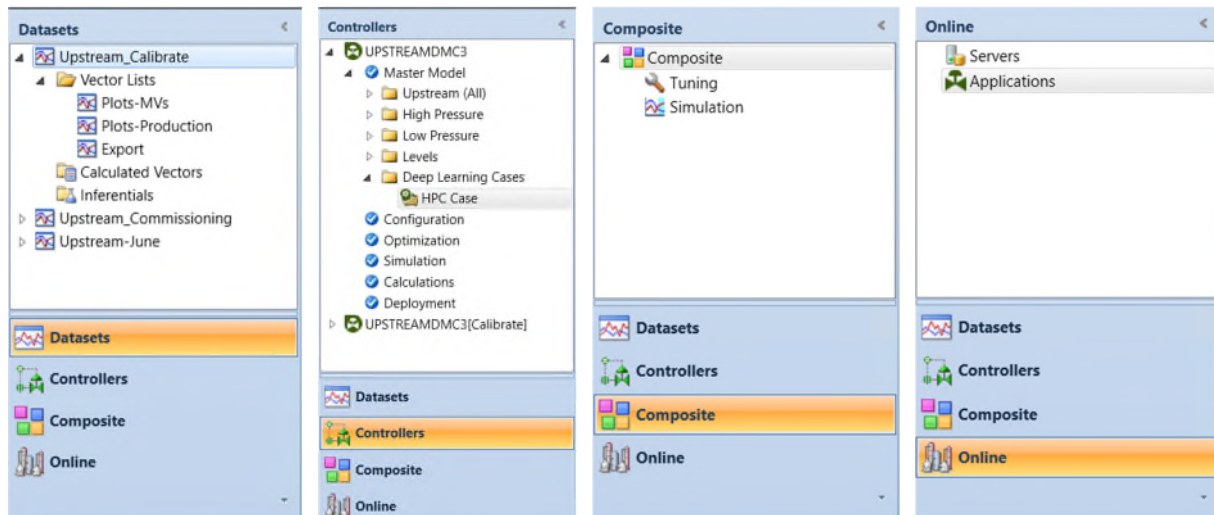
Single Environment to build, configure and deploy controllers.



Interface Layout

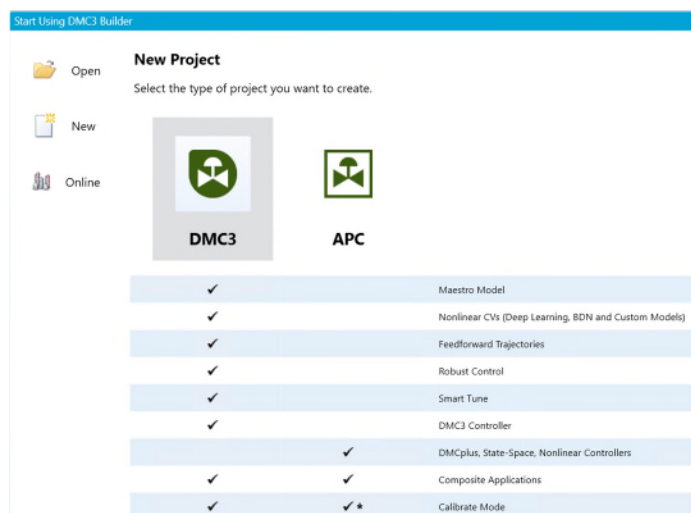


Navigation Tree Workflow



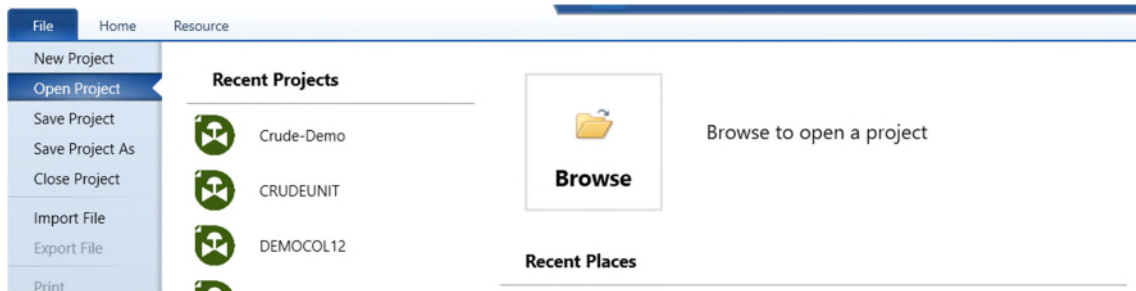
Creating a Controller Project (1)

- Step 1: Navigate to **File | New Project**

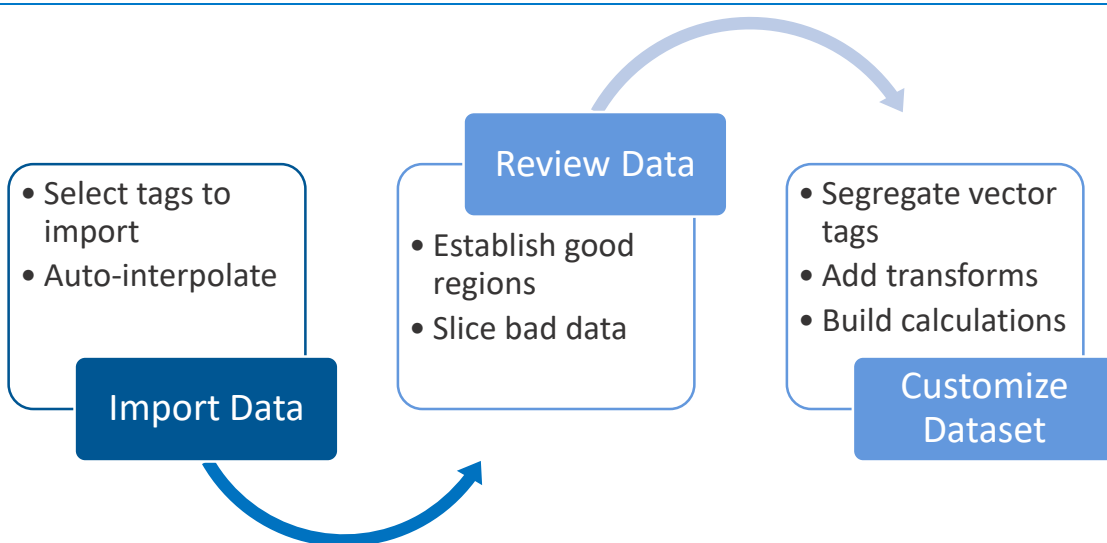


Loading an Existing Project

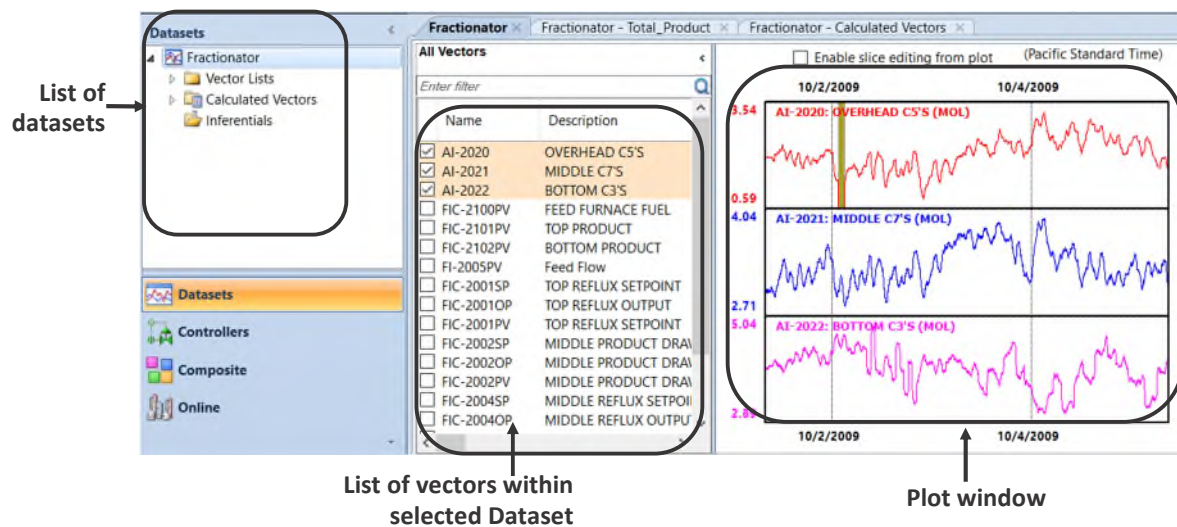
- To load an existing project use the **File | Open Project** from the main menu



Dataset Workflow



Reviewing Dataset



Slicing Options: Mark Bad or Interpolated

Data to slice:

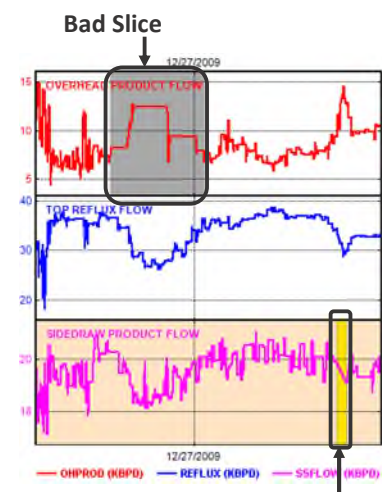
- Communication failures
- Instrument failures
- Inconsistent PID performance (spikes, saturation, etc.)
- Unmeasured process disturbances
- Closed loop feedback

Exclude data from modeling

Change slicing selection



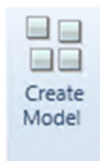
Replace bad/missing data by interpolating end points



Interpolated Slice

Model Identification (1)

- Click Create Model from top tools ribbon to use the dataset to create the controller



Model Properties
(TTSS, Execution Frequency)

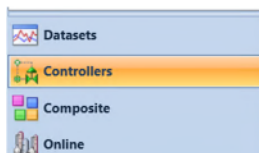
Build Controller Structure
(Independents & Dependents)

The dialog box 'Identify Model - Specify Structure' contains the following elements:

- Properties:** Name (NewModel), Sample Period (1 Minutes), Time to Steady State (120 Minutes).
- Model Type:** Radio buttons for 'Empty Model' and 'Select Variables from a Dataset' (selected).
- Variables:** Dataset (Fractionator), List (All Vectors), Filter.
- Table:** A table with columns: Name, Tag Name, Input, Output, Units, Description. It lists various process variables like AI-2020, FIC-2100PV, etc.
- Buttons:** Add, Delete, Move Up, Move Down, Check, Uncheck, OK, Cancel, Help.

Model Identification (2)

- Creating the model will take you from the Datasets to Controllers view



Master Model

Models Tool Ribbon

The 'Models Tool Ribbon' includes buttons for Import, New Controller, Add, Import Case, Merge Model, Up, Down, Master Model, Constraints, Summary, Edit Variables, Transforms, Identify All, and Predict All. Below it, the 'Master Model Structure' dialog box shows a table for defining model structure.

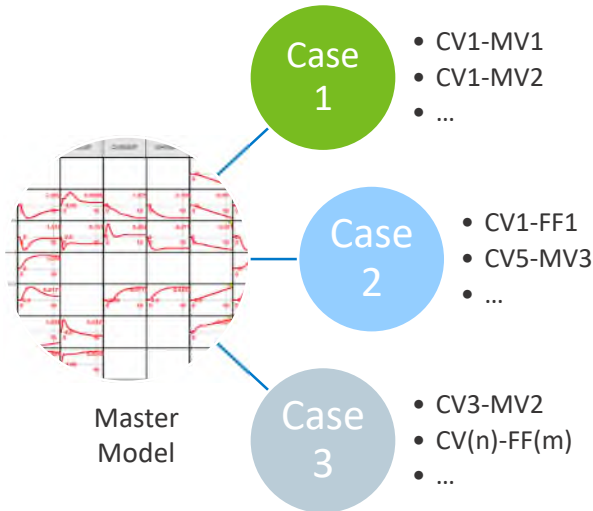
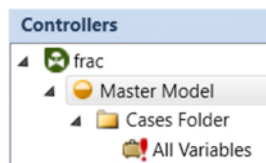
Output	Typical Move	DRAWT	COLDP
OHPROD	1		
REFLUX	1		

ID Cases

Master Model Structure

Master Model

- Run trials for model identification with varying configuration settings in different Cases
- Then pick and choose curves from Cases to build the final Master Model for use online



Case | Variables View

- Case Variables section is used to define dataset and input / output variables for model identification

Select & Modify Dataset

Customize & Modify Vectors

Independent Variables

Dependent Variables

Case | Parameter Trials View

- Case Parameter Trials section is used to set up FIR and Subspace identification parameters
- Only one trial can be selected for each of the model analysis tools checkboxes (more on this later):
 - Prediction
 - Frequency Uncertainty plot
 - Time Uncertainty plot

Subspace Model ID Parameters

Time to Steady State	Oversample	Max Order	Prediction	Uncertainty	Time Uncertainty
90	1	30	☒	☐	☐
120	2	30	☐	☐	☐

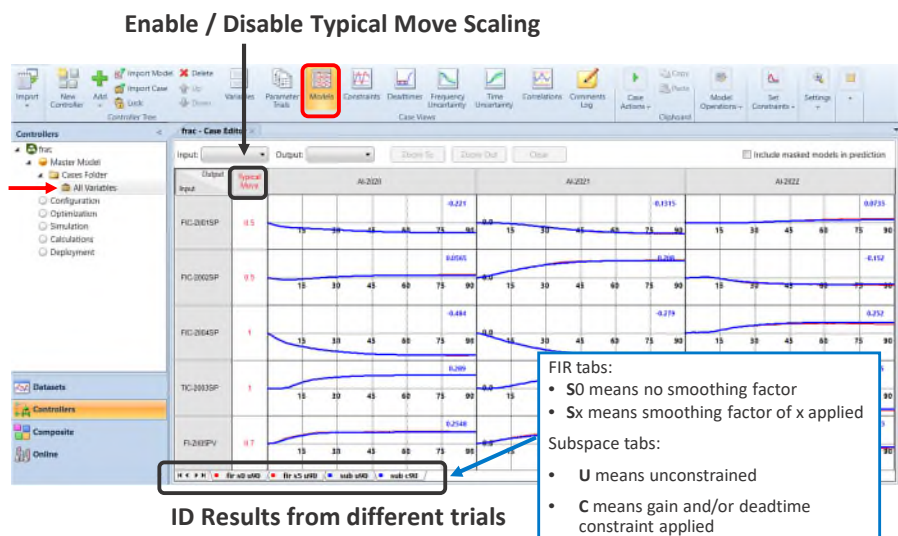
Add/Remove ID Trials

FIR Model ID Parameters

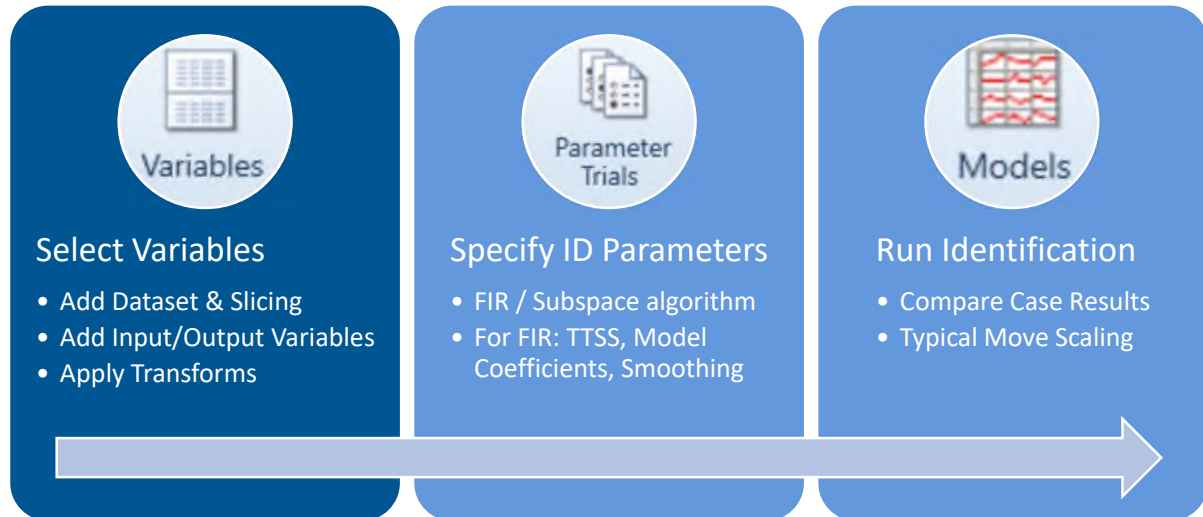
Time to Steady State	Coefficients	Smoothing	Prediction	Uncertainty	Time Uncertainty
30	30	5	☐	☐	☐
60	60	5	☐	☐	☐
90	90	5	☐	☐	☐

Models View - Identification Results

- Models view shows model ID results
- Typical Move scaling can be enabled by clicking on the column header and it will appear as **Red** text
- Note that each parameter trial has two tabs for comparison



Case Modeling Workflow



Model Identification: Mini-Workshop

DMC3 Builder Workflow

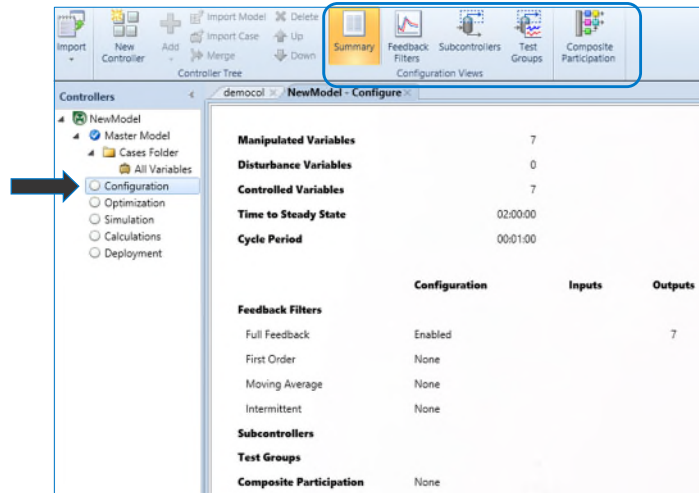
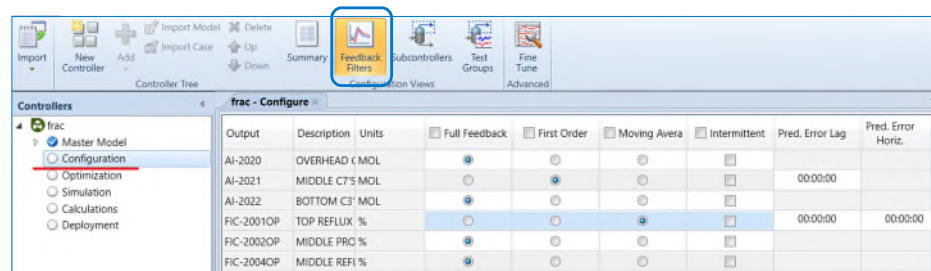


Figure: Screenshot of DMC3 Builder Controllers navigation tree where Master Model is complete (Blue checkmark) so we can move on to Configuration

DMC3 Builder: Filter Configuration

- Prediction Error Filters are enabled in the Configuration section of DMC3 Builder controller navigation tree
- This is also where a variable can be set as intermittent if required



Full Feedback is no filter (default)

Select type of prediction error filter

Select if intermittent measurement

DMC3 Builder Workflow

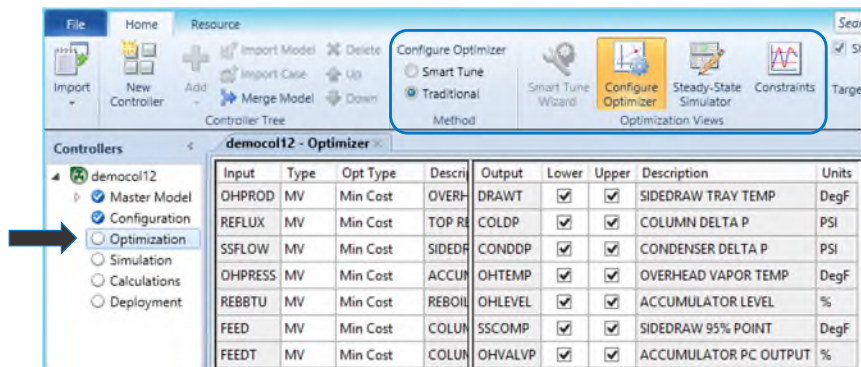


Figure: Screenshot of DMC3 Builder Controllers navigation tree, Optimization section, where the Steady State Optimizer tuning is configured

DMC3 Builder Workflow

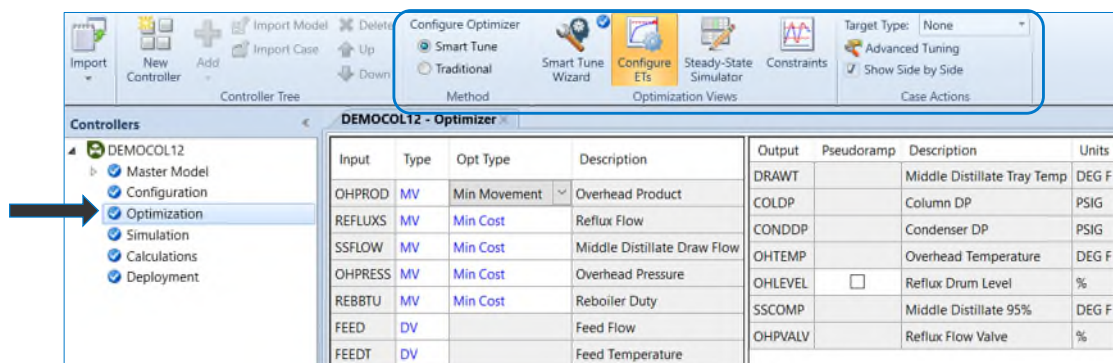


Figure: Screenshot of DMC3 Builder Controllers navigation tree, Optimization section, where the Steady State Optimizer tuning is configured

DMC3 Builder: Optimizer Configuration

Enter operating conditions

Initialize tuning with data

Inputs	Combined Status	Service Request	Measurement	Validity Low Limit	Engineering Low Limit	Operator Low Limit	Steady State Value	Ideal Stead State	
OHPROD	Normal	On	0.2	0	5.3	6.3	0.2	0	
REFLUX	Normal	On	35	0	25	27	35	0	Normal
SSFLOW	Normal	On	20	0	18	18	20	0	Normal
OHPPRESS	Normal	On	16	0	13	14	16	0	Normal
REBBTU	Normal	On	2100	0	1400	1500	2100	0	Normal
FEED	Normal	On	39	0			39	0	Normal

Outputs	Ideal Constraint	SS Current Move	Operator High Limit	Engineering High Limit	Validity High Limit	LP Cost	SS Low Concern	SS Low Rank	SS High Concern	SS High Rank	Critical
COLDP	Normal	0	3.8	3.9	1E+10	0	2	310	2	310	No
CONDDP	Normal	0	22	23	1E+10	0	0.2	900	0.2	250	No
OHTEMP	Normal	0	130	130	1E+10	0	0.2	900	0.2	260	No
OHLEVEL	Normal	0	67	73	1E+10	0	3	300	3	320	No
SSCOMP	Normal	0	360	360	1E+10	0	5	100	5	100	No
OHVALVP	Normal	0	59	61	1E+10	0	2	320	2	320	No

SmartTune LP Strategy Page

LP Strategy shows the order and direction in which MVs will be moved when a new CV constraint becomes active. You can choose MV, CV, and subcontroller priorities to evaluate the impact on CV constraint handling. View the costs calculated by the Smart Tune engine using the Calculate Costs button. Save changes using the Apply button.

Subcontroller: All Variables Show Costs

Indicators: Positive gain (blue), Negative gain (red), Ignored/Tradeoff (grey), Order of MV move (1-6)

Movement direction: Solid when up/down direction favors the economic objective, Up/down movement direction (blue/red arrows)

Example: For low limit constraint, MV will move up, 2nd in order, with positive gain that favors the economic objective.

Variables	Priorities Original / New	DELTA_TEMP	DELTA_FLOW	FEED_TARGET	FEED_PRES_LPV	FEED_PRES_ZPV	REFLUX_FLPV	REFLUX_RPV	REFLUX_TPV	FEED_PRES_FPV	JET_ST_LVGP	LD_PRES_FPV	LD_ST_LVGP	TP_CIR_RPV	HD_ST_LVGP	HD_PRES_FPV	BT_CIR_RPV	STM_PRES_FPV
HEATER	1 1																	
FEED_1.SP	1 1																	
FEED_2.SP	1 1																	
FUR_COT_1.SP	2 2																	
FUR_COT_2.SP	2 2																	
COLUMN	2 2																	
OVH_PRESS.SP	5 5																	
OVH_TEMP.SP																		
JET_FLOW.SP	9 9																	
LD_FLOW.SP	7 7																	
TP_CIR_FLPV																		
TP_CIR_TPV																		

Apply Reset Recalculate Costs

DMC3 Builder Workflow

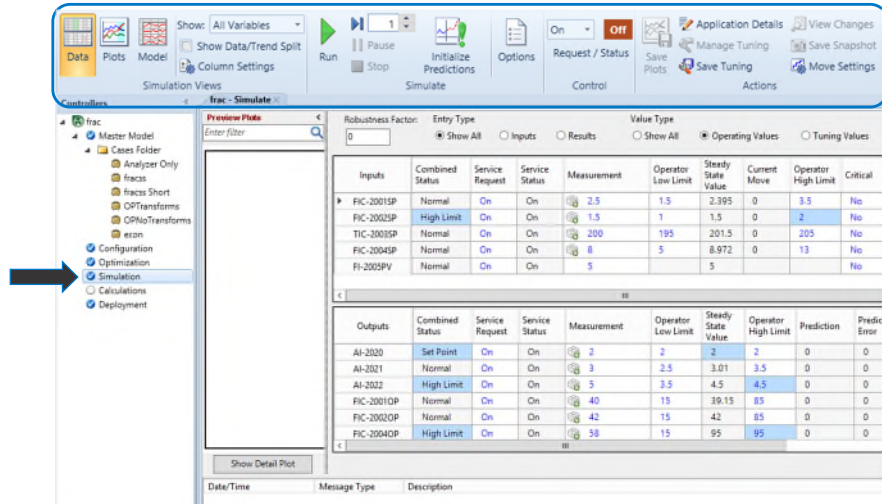


Figure: Screenshot of DMC3 Builder Controllers navigation tree, Simulation section, where the dynamic tuning is configured

DMC3 Builder Workflow

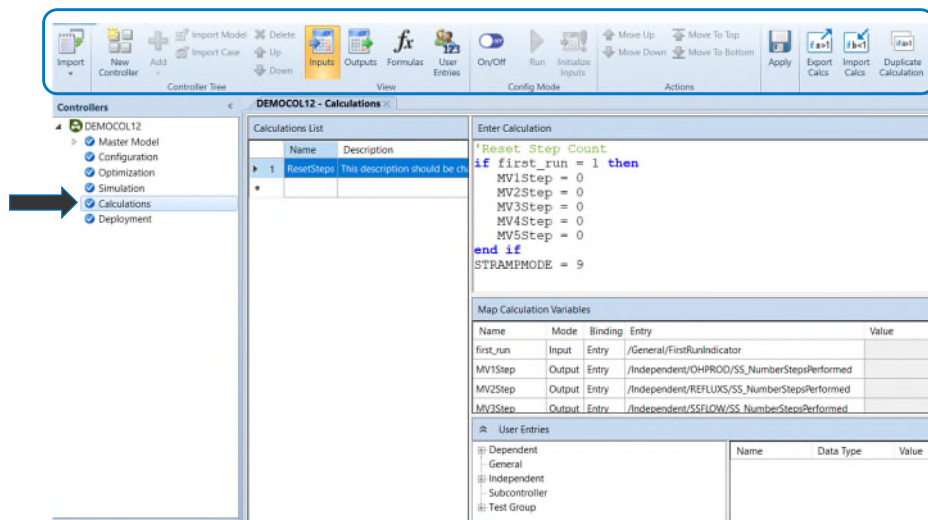


Figure: Screenshot of DMC3 Builder Controllers navigation tree, Calculations section

DMC3 Builder Workflow

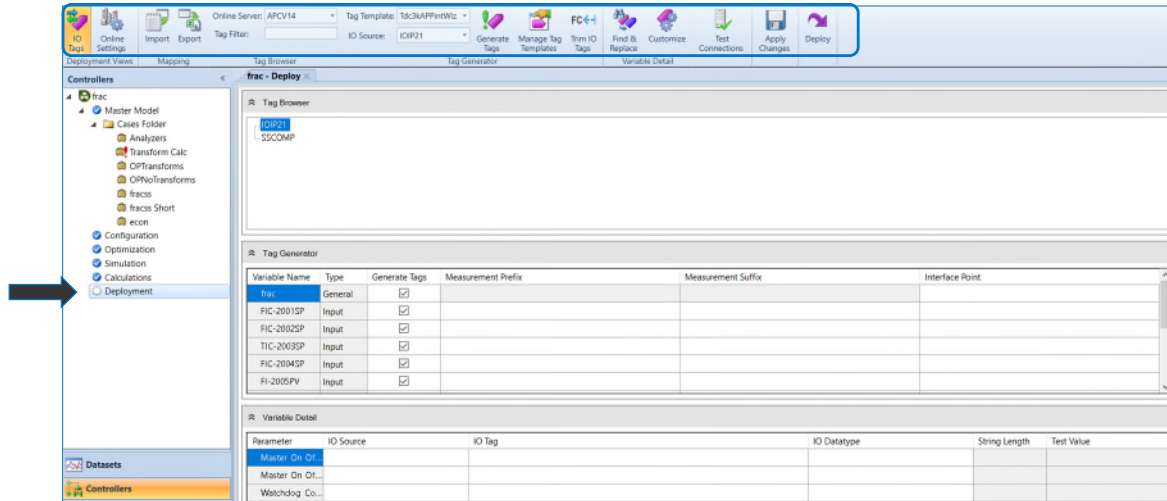
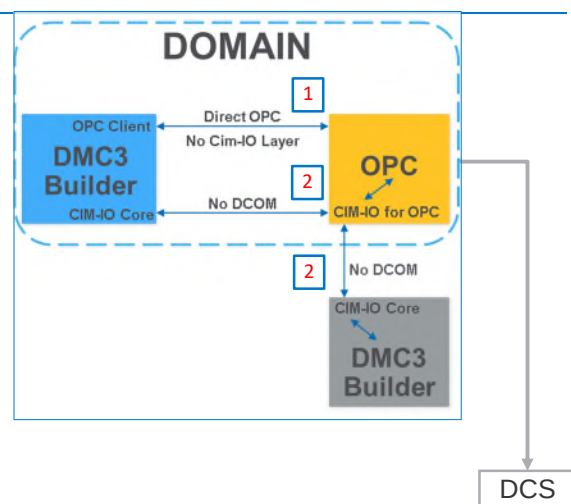


Figure: Screenshot of DMC3 Builder Controllers navigation tree, Deployment section to connect controller parameters to DCS tags

DMC3 Builder: Configure Connection

Two ways to connect DMC3 to a DCS:

1. DMC3 as a direct OPC Client
 - DCS and DMC3 must run in the same domain
 - Use DMC3 Builder's built-in OPC Client for direct connection to an OPC server without the need for intermediate CIM-IO servers
2. DMC3 as a CIMIO Client
 - DCS and DMC3 can run in the same or different domains
 - Connect to the DCS through CIM-IO for OPC interface
 - CIM-IO server can be either on the APC Online server (requires DCOM permissions) or on the OPC server (bypasses DCOM)
 - Most common method and best practice



Configure Online Server Program

- Stand-alone program on the APC Online Server
- Includes settings for the online server and applications:
 - History retention
 - Automated snapshots
 - Users and roles
 - Permissions
 - IO Sources for direct OPC and/or CIMIO connection
 - Diagnostic logging

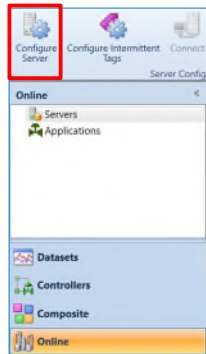


Figure: DMC3 Builder / Online section has a button to open Configure Online Server; this can also be opened directly from Windows Start menu

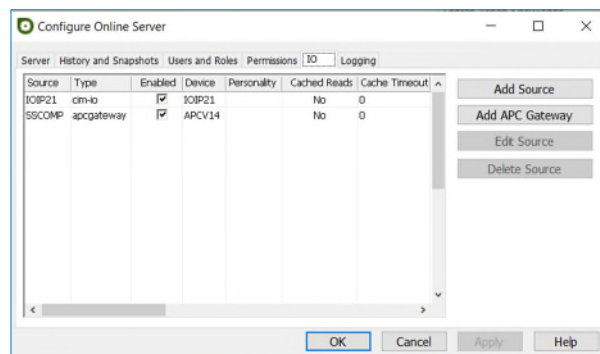


Figure: Configure Online Server program / IO tab is used to create IO Sources that receive data from DCS either through OPC or CIMIO connection, or the APC Gateway to receive data from other APC applications

DMC3 Builder: Deployment Workflow

Select IO connection data source

Configure parameter list requiring IO

Select Tag generation template

Generate tag connections/mapping to the DCS

Test Connection and prepare to deploy application

DMC3 Builder: Configure Connection

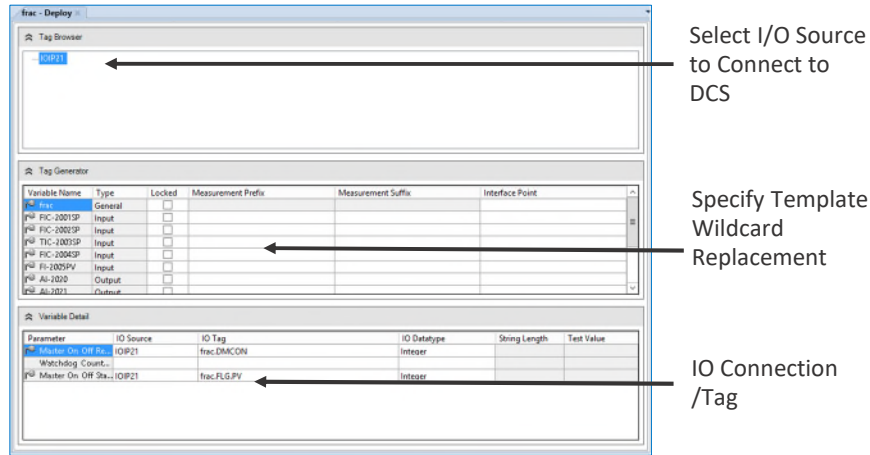


Figure: DMC3 Builder | Controller | Deployment section for IO Tags

Configure Connection: Customize IO

- Customize which controller parameters should be connected to an IO tag.

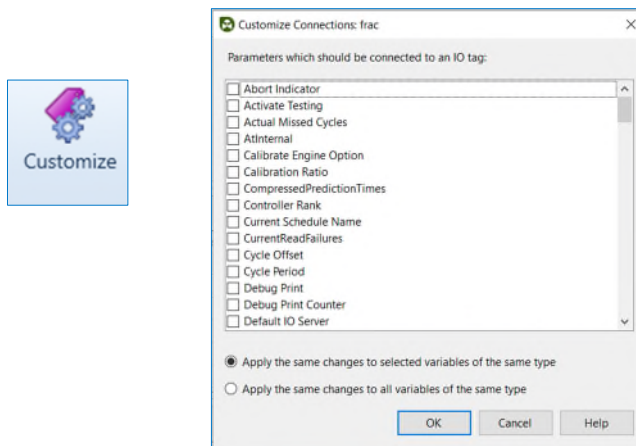
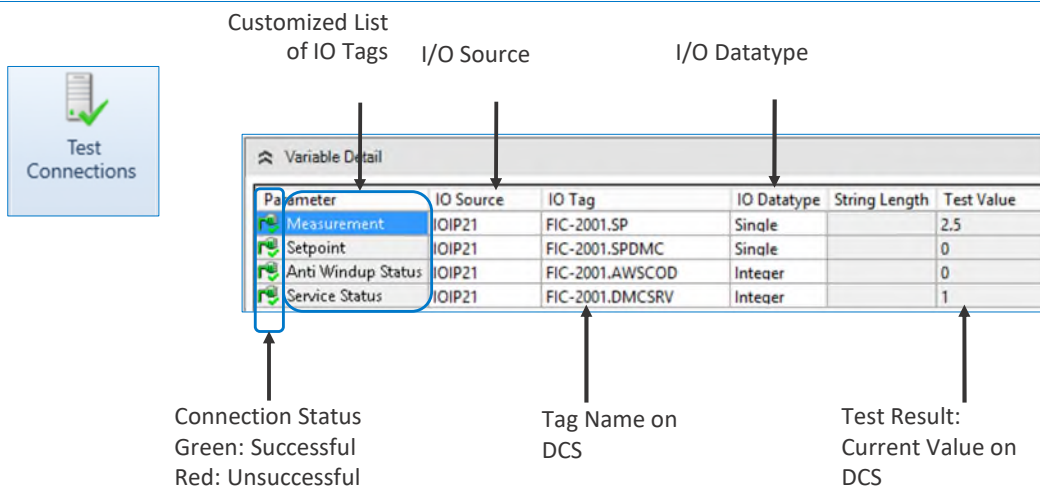


Figure: DMC3 Builder | Controller | Deployment section Tools Ribbon button for Customize

Configure Connection: Variable Details

Customized List of IO Tags I/O Source I/O Datatype



The 'Test Connections' button is shown on the left. The 'Variable Detail' table is shown on the right, with columns: Parameter, IO Source, IO Tag, IO Datatype, String Length, and Test Value. The table contains four rows: Measurement, Setpoint, Anti Windup Status, and Service Status. Arrows point from the labels above to the corresponding columns in the table. A blue box highlights the 'Parameter' column, and a green box highlights the 'Service Status' row. Arrows point from the labels below to the corresponding cells in the table.

Parameter	IO Source	IO Tag	IO Datatype	String Length	Test Value
Measurement	IOIP21	FIC-2001.SP	Single		2.5
Setpoint	IOIP21	FIC-2001.SPDMC	Single		0
Anti Windup Status	IOIP21	FIC-2001.AWSCOD	Integer		0
Service Status	IOIP21	FIC-2001.DMCSRV	Integer		1

Connection Status
Green: Successful
Red: Unsuccessful

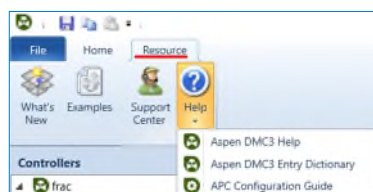
Tag Name on DCS

Test Result:
Current Value on DCS

Figure: DMC3 Builder | Controller | Deployment | Variable Detail section after hitting Test Connections

DMC3 Help Files

- There are three Help files provided in DMC3 Builder | Resource tab | Help section:
 - DMC3 Help** – information about using the **DMC3 Builder** desktop tool (dialog windows, buttons, configuration of controllers, etc)
 - DMC3 Entry Dictionary** – list of all built-in entries used by the controller (such as status, request, limit, etc), along with explanation of possible values that can be set for that entry
 - APC Configuration Guide** – procedure to set up the APC servers after installation is complete



Controller Deployment work flow: Demo



Aspen DMC3 Additional Features from DMCplus



Lesson Objectives

- In this session you will learn about:
 - DMCplus vs DMC3 (ACO vs RTE)
 - DMC3 features and benefits
 - DMCplus to DMC3 conversion (DEMO)

AspenTech Tools Terminology: ACO vs RTE

- AspenTech provides 2 platforms to build models, configure controllers, run offline simulation, and deploy DMCplus/DMC3 controllers to run online
 1. **ACO** (Advanced Control & Optimization) – Refers to using legacy tools including DMCplus Model, DMCplus Build, DMCplus Simulate and APCManage
 2. **RTE** (Real-Time Environment) – Refers to using the DMC3 Builder tool
- In both platforms, the user can build a DMC3 controller (traditional linear control solution) or a DMC3 controller (traditional solution plus advanced features like Calibrate, Adaptive Modelling, etc.)
 - Note: AspenTech announced End of Life schedule for DMCplus (MDL) controllers starting 2023, V14 is the last version supported, but the ACO platform tools are still available for DMC3 controllers
 - Lifecycle Announcement: [KB 100803](#)
 - More info on ACO vs RTE: [KB 99492](#)

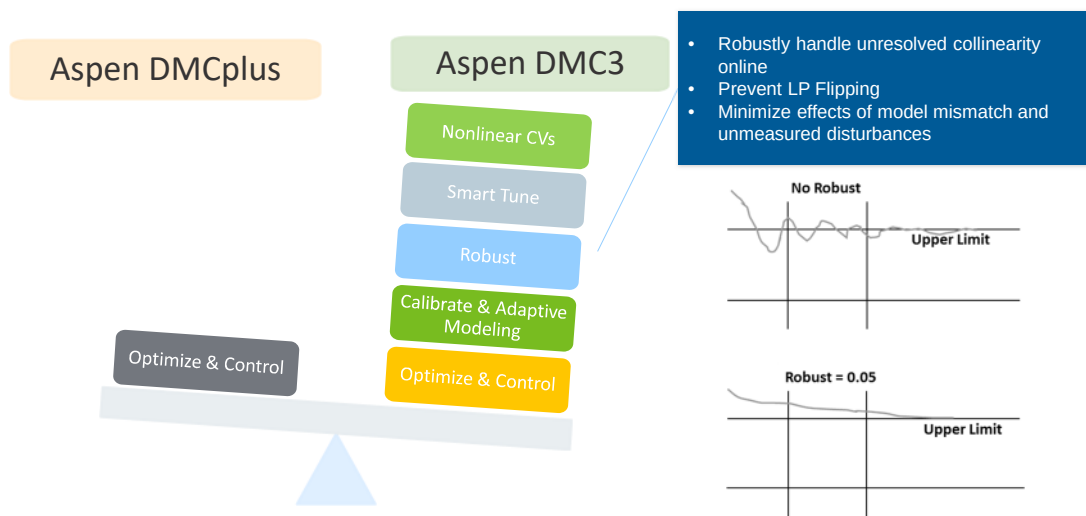
Steps to **Build** and Configure DMCplus/DMC3 Controller – **Offline**

	ACO-Based Platform Tools	RTE (DMC3 Builder) Platform Node
Build the Model	DMCplus Model (export .mdl or .mdl3 file)	DMC3 Builder – Master Model
Configure Steady State Tuning	DMCplus Build (.ccf file)	DMC3 Builder – Configuration, Optimization
Configure Dynamics Tuning	DMCplus Build	DMC3 Builder – Simulation
Configure Custom Calculations	DMCplus Build	DMC3 Builder – Calculations
Simulate controller to verify tunings	DMCplus Simulate	DMC3 Builder – Simulation
Map to DCS Tags	DMCplus Build	DMC3 Builder – Deployment

Steps to **Deploy** an DMCplus/DMC3 Controller – **Online**

	ACO-Based Platform Tools	RTE Platform Node
Deploy the Controller to Online Server	1. Copy .ccf and .mdl/.mdl3 files to the appropriate folder on the online server (ProgramData\AspenTech\APC\Online\app) 2. Use PCWS or APCMANAGE tool to load the controller	Connect to the Online Server and Deploy from DMC3 Builder
Start the Controller	Use PCWS or APCMANAGE to start/run the controller	Use PCWS or DMC3 Builder to start/run the controller
Enable Aspen Watch (AW) to collect controller data	1. Copy .ccf and .mdl/.mdl3 files to the appropriate folder on the AW server (ProgramData\AspenTech\APC\Performance Monitor\app\) 2. Use Watch Maker to add the controller 3. Use PCWS or Watch Maker to enable monitoring	Use PCWS or Watch Maker to enable monitoring

Aspen DMC3: Robust Control



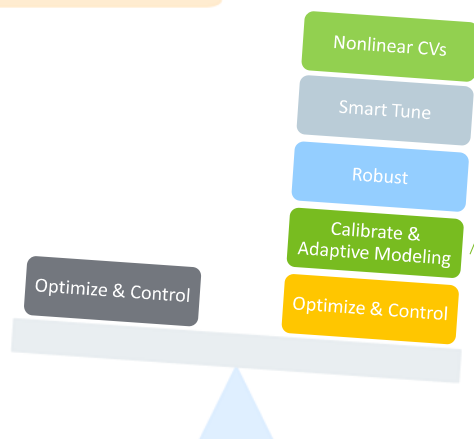
Additional Resources on Robust Control

- The Aspen DMC3 Difference – Whitepaper
 - https://esupport.aspentech.com/S_Article?id=000065940
- White Paper: DMC3 Robust Control
 - https://esupport.aspentech.com/S_Article?id=000101033

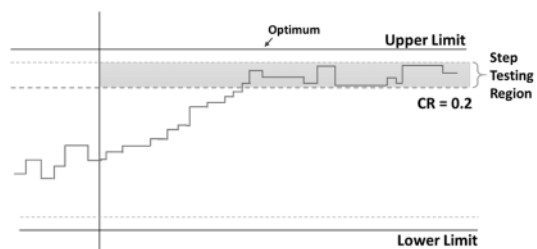
Aspen DMC3: Calibrate & Adaptive

Aspen DMCplus

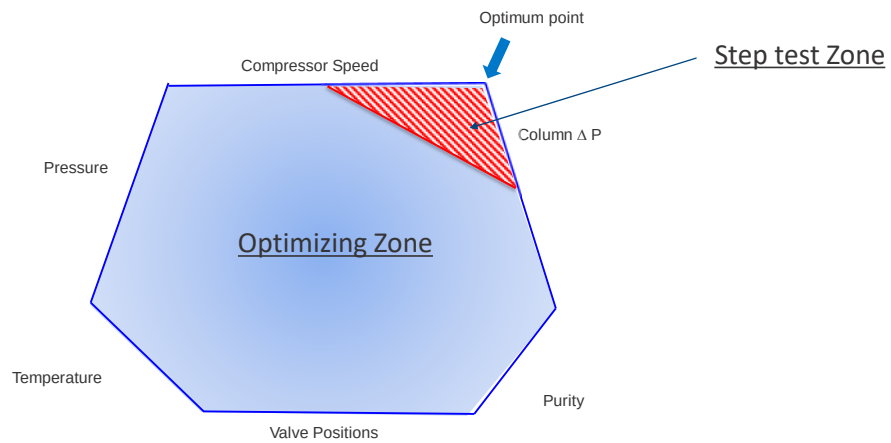
Aspen DMC3



- Step test while accruing benefits
- Make smaller discrete moves
- Update controller models online



Aspen DMC3: Calibrate & Adaptive

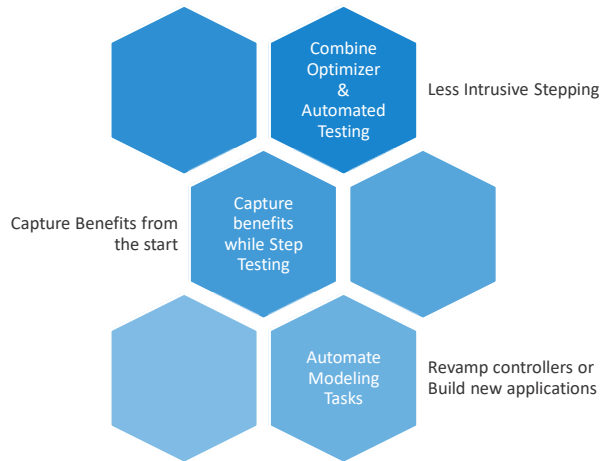


Reference KB about Calibrate

Calibrate 2.1 Engine in APC V12

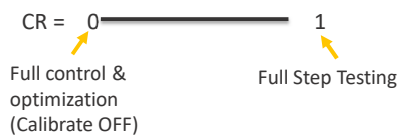
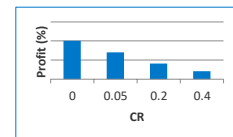
https://esupport.aspentech.com/S_Article?id=000098170

Calibrate Mode & Adaptive Modelling



Calibrate Ratio parameter divides the operating area into two sub-areas:

- Optimizing
- Automated Step-testing



- Learn more about Robust & Calibrate in **APC250** class

Aspen Adaptive Modeling in PCWS

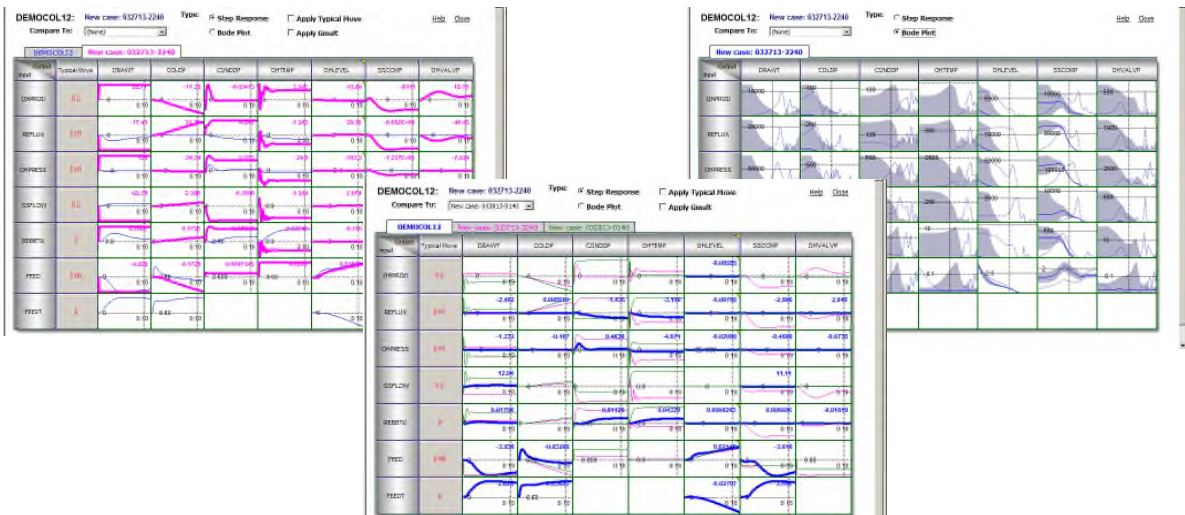
- The Modeling page can be accessed through **PCWS | Online | Model | Modeling**
- Two type of cases, model identification (ID) and model quality (MQA)

Case Name	Status	Next Schedule	Data Start	Data End	Actions
AUTOID	Failed	None	01-FEB-13 15:35:12	19-MAR-13 21:55:48	[Icons]
AUTOSLICE	Failed	None	19-MAR-13 19:25:48	19-MAR-13 23:25:48	[Icons]
New case	Scheduled (remaining: 2)	28-MAR-13 04:40:25.0			[Icons]
032713-2240	Complete		27-MAR-13 14:35:25	27-MAR-13 22:40:25	[Icons]
032813-0140	Complete		27-MAR-13 14:35:25	28-MAR-13 01:40:25	[Icons]

Case Name	Status	Next Schedule	Data Start	Data End	Actions
LabExercise	Complete	None	25-FEB-13 08:52:25	25-FEB-13 10:52:25	[Icons]
qual chck1	Complete	None	26-FEB-13 12:02:57	26-FEB-13 15:08:57	[Icons]
021413-0208	Complete		13-FEB-13 15:02:57	14-FEB-13 02:07:57	[Icons]

- Run
- Edit
- Delete
- Log file
- Raw data
- Result
- Assemble model
- Download model

Model Viewer



DMC3 additional features: Quick Demo

Conversion from DMCplus to DMC3

How to Convert a Legacy Aspen DMCplus (ACO) Application to Aspen DMC3?

https://esupport.aspentech.com/S_Article?id=000094882



Conversion from DMCplus to DMC3: Demo



Future Trends in Process Control



Lesson Objectives

Upon completion of this lesson, you will be able to identify:

- The role of AI/ML in advanced process control
- Aspen Maestro Model
- Aspen Deep Learning for DMC3 and AspenIQ

AspenTech Providing AI to Industry

AspenTech, Inc. acquired AIoT Company **mnubo** in July 2019

Aspen's AI solutions combine Engineering/Domain knowledge and AI/ML technology

Engineering Fundamentals
Domain Expertise

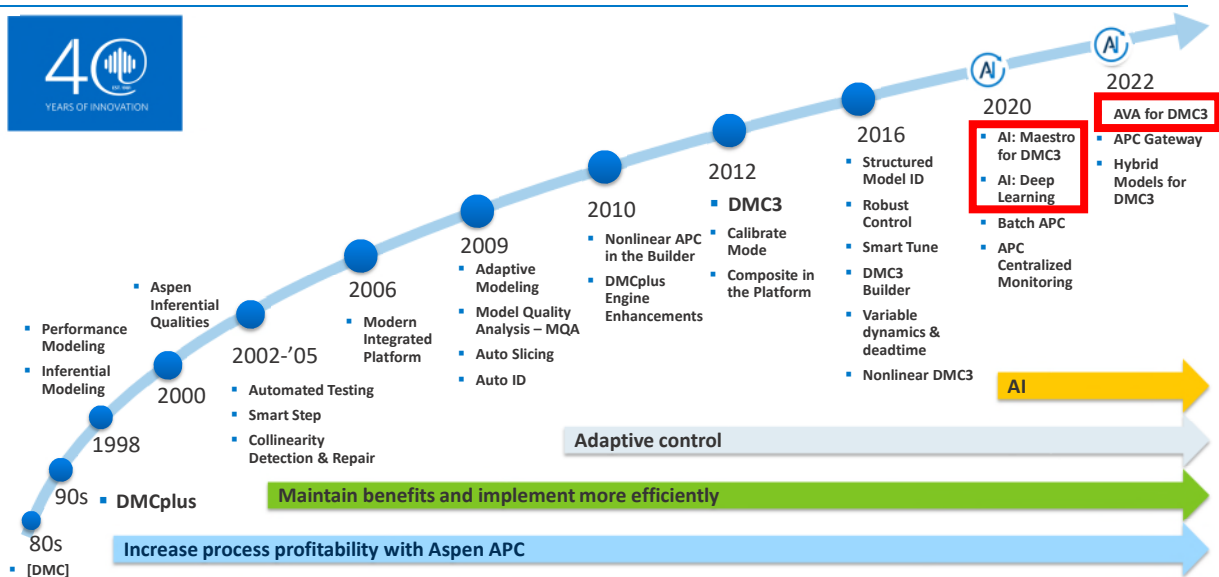


AI/ML
Advanced Analytics

Industrial AI

Insight | Guidance | Automation

Aspen APC History



AI/ML related new features in Aspen DMC3

Maestro for DMC3

- Utilize historical data to build seed models (needed by Calibrate) via a self-guided workflow
- Capitalize on parallel computing to process large datasets and multiple cases for model-building

Deep learning for DMC3 and IQ

- Nonlinear deep learning neural network Modeling is available for DMC3 CV and IQ modeling

Aspen Virtual Advisor(AVA) for DMC3

- *Elevates human-machine interactions with a 24X7 AI virtual advisor*
- *Users find answers to common process control questions and get actionable guidance*

What is Maestro Model?

- Maestro Model is a simple, guided workflow incorporated into Aspen DMC3 Builder starting in V12.0 that facilitates the creation of seed models from historical data
- Seed models are needed prior to applying Calibrate to generate step-test data
- Maestro leverages various machine learning techniques, including:
 - Data mining
 - Auto-slicing
 - Input correlation
 - Transformation detection
 - Output analytics

Maestro Model Capabilities

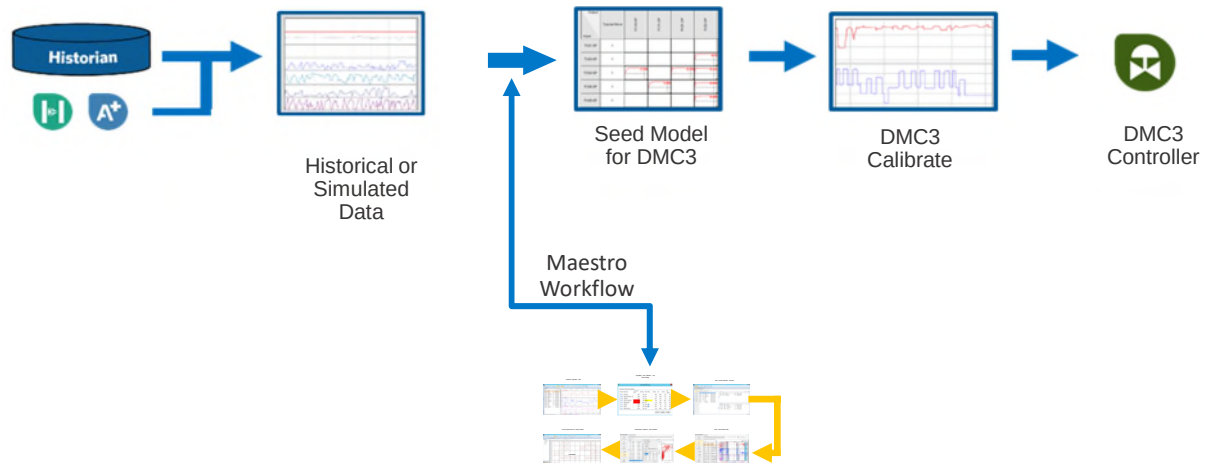
Maestro has revolutionized the APC model building process - how?

- AI technology and parallel computing dramatically reduce model-building complexity & time
- Powerful Machine Learning algorithms are used to mine the wealth of historical process data to build seed models
- Model-building expertise is packaged in a simple, self-guided workflow
- Seed models are easily deployed online as self-learning DMC3 applications

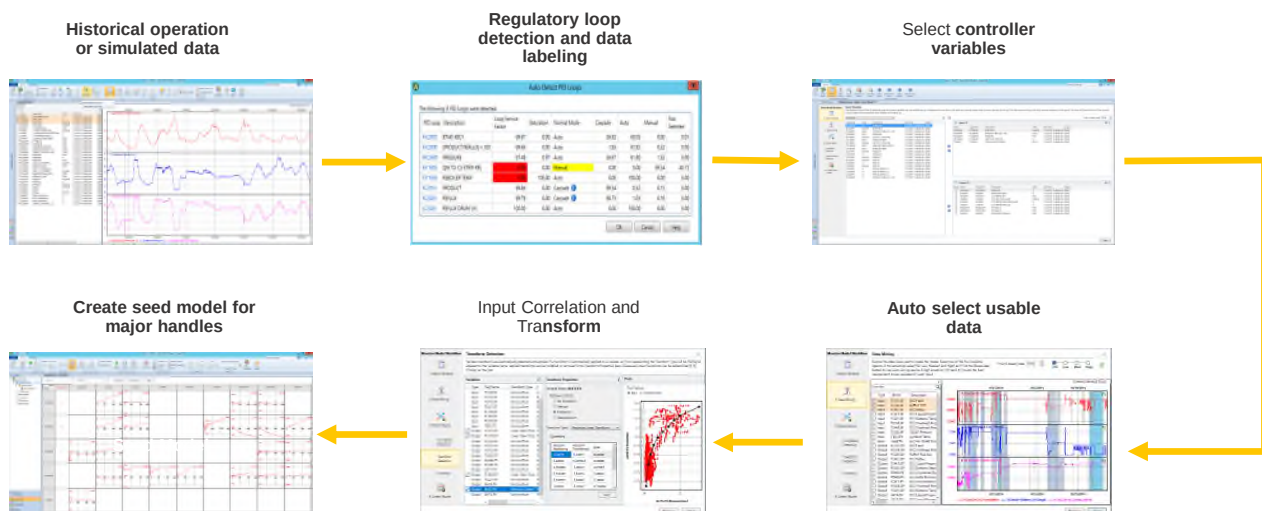
Purpose of Maestro

- The main purpose of this feature is to help users speed up building seed models from historical data and begin realizing the benefits from Calibrate technology
- Maestro is capable of processing large amounts of data and finding useful moves for the manipulated variables that can yield good seed models
- It also includes embedded APC knowledge in an easy-to-use workflow that provides guidance at every step

APC Project Workflow with Maestro



Maestro Model Scheme

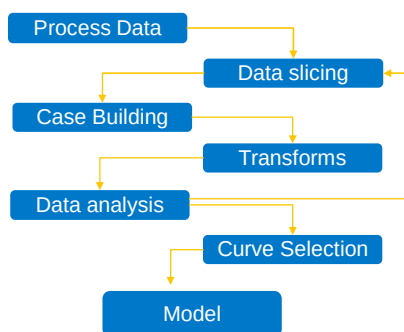


Benefits of Maestro Model

- Data mining AI machine learning algorithms support a guided automated workflow, which previously required a control expert
- Drastically reduce engineering effort & time to build and deploy APC models
- Build models remotely
- Expand APC benefits in new industries with limited prior APC footprint
- Lower the barrier of DMC3 implementation
- Make the DMC3 model building process more accessible to new users

Traditional Approach for Building the Seed Model

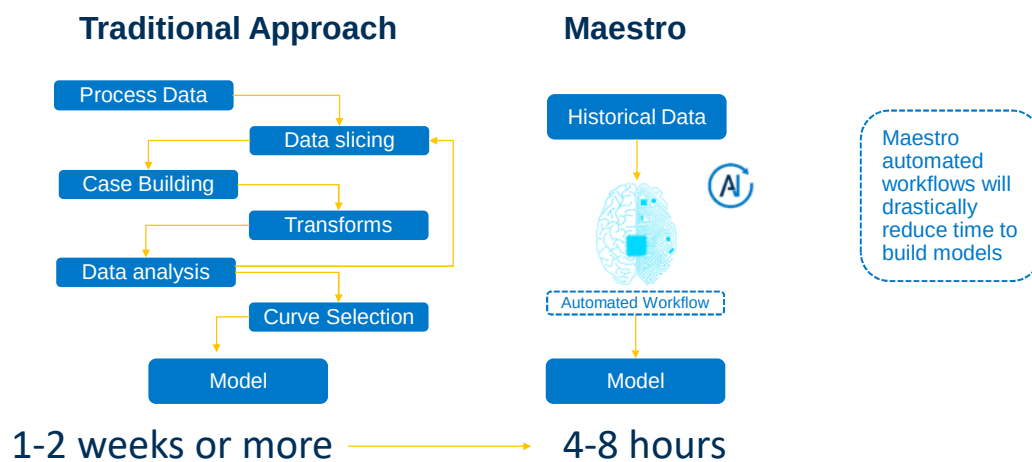
Traditional Approach



1-2 weeks or more

- Traditional approach is an iterative process that requires the use of step test data to create an initial seed model
- The timeline from collecting data to creating a model can be 1 to 2 weeks or more, depending on:
 - The quality of manual or automated step test data needed for identification (need clear steps, open-loop data, etc.)
 - The expertise of APC engineer(s)
 - Available time, resources and funds for the project
 - Requirements for model quality
 - Ability to extract step test data from historical data, a labor intensive and time-consuming task

Compare Traditional Approach to Maestro



Link to Patent

More information can be found in the patent:

System And Methods For Automated Model Development From Plant Historical Data For Advanced Process Control: <https://patents.google.com/patent/US20210116891A1/>

More information is available about Neural networks below:

https://www.youtube.com/watch?v=oJNHXPsoXDk&ab_channel=SteveBrunton

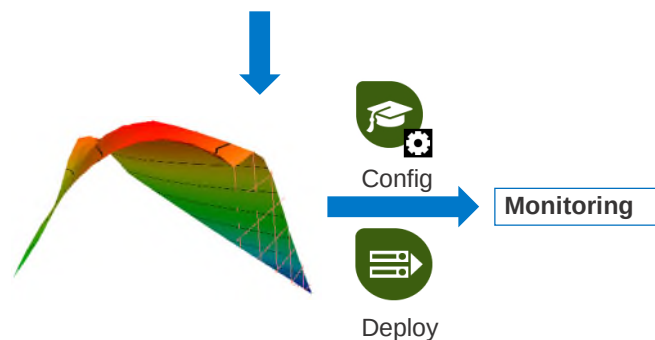
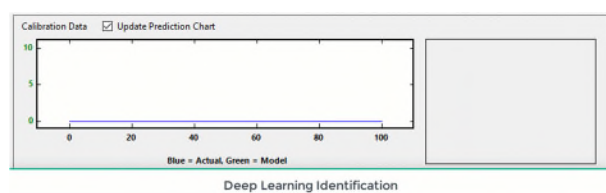
This video has variety of neural network architectures available to solve various problems in science and engineering.

Maestro Model: Quick Demo on Overall Workflow

Aspen Deep Learning – For Inferential Quality Models

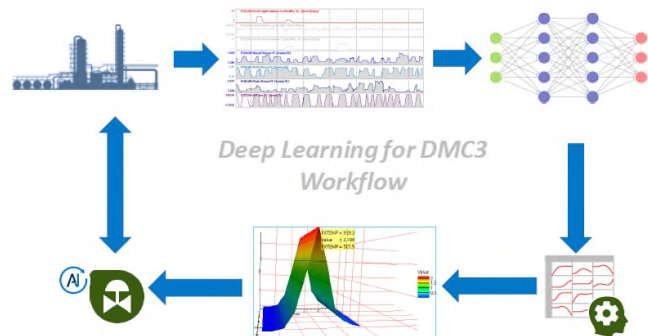
Aspen IQ with AI Capabilities

- Deep Learning extends the applicability of AspenIQ to non-linear processes
- Build Deep Learning models for Aspen IQ using DMC3 Builder workflow
- Deploy Deep Learning IQ inferentials using the current Aspen IQ software, which includes:
 - Aspen IQ Config (Configuration)
 - APCManage (Deployment)



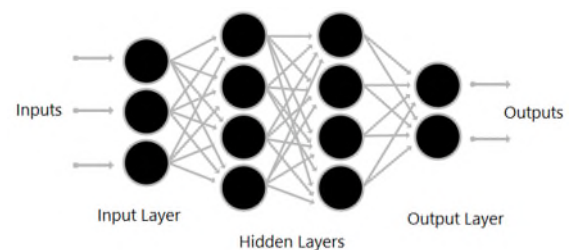
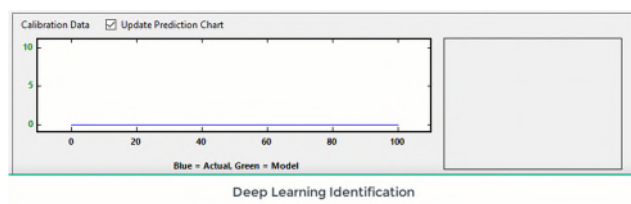
Aspen Deep Learning for DMC3

- Deep Learning extends the applicability of DMC3 to non-linear processes
- New workflow for building Deep Learning Models
 - Deep Learning CV can be added to existing or new DMC3 application
- Can now add Constraints –
 - Gain Ratio and General Gain Constraints
 - Deep Learning constraints can be utilized with Linear models.
- Simulation and Gain Analysis in Aspen DMC3 Builder
 - Simulate the DL model in DMC3 Builder simulation



How DL model identification works

- End-to-end learning is achieved with Deep Learning (DL)
- The term “deep” is related to the number of hidden layers in a neural network
- DL model learns features from the dataset
- Available types of deep neural networks:
 - GRU (Gated Recurrent Unit) For Dynamic Model
 - LSTM (Long Short-Term Memory) For Dynamic Model
 - MLP (Multilayer perceptron) For Steady state Model



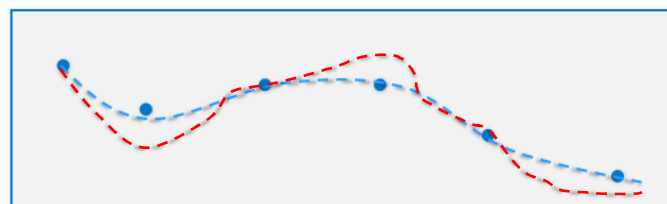
Capabilities of Deep Learning for DMC3 and IQ

- **Deep Learning for DMC3 is capable of:**
 - Integrating with powerful 3rd party ML model-building package TensorFlow (open-source library for neural networks-ML applications)
 - Augmenting model predictions with new data (preserve model memory through incremental data)
- **Deep Learning for DMC3 consist of:**
 - Combination of mature Aspen IQ technology with DL and Deep Learning for DMC3
 - More accurate models with DL
 - Install base and customers/services ready to deploy
 - Constraints for online controller

Capabilities of Deep Learning (Cont..)

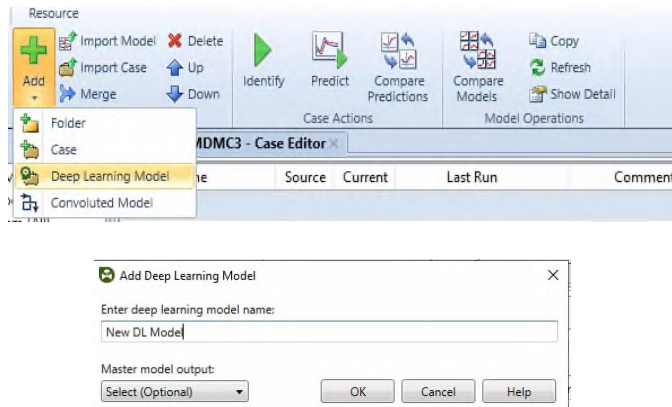
- Easy to build Deep Learning models in DMC3 Builder
- Deep Learning algorithms available to model steady-state and dynamic data
- Aspen Watch monitoring included
- Easy deployment with well-known Aspen IQ Config and APCManage

Provide a powerful framework for online deployment and monitoring of DL models



- Plant measurement
- - - Standard approach (performance degradation)
- - - Deep Learning (linger sustainability)

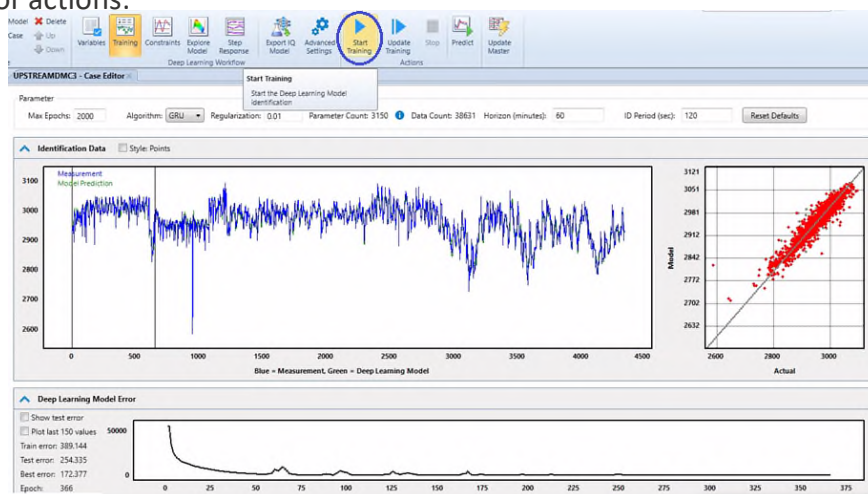
Deep Learning Cases



- Deep Learning models can be created from case editor section
- Use the add button to create a new Deep Learning case
- The following dialog box to add Deep Learning model appears on selecting the “Deep Learning Model” selection from Cases section

Deep Learning Model Identification

The Deep Learning identification dialog is accessed from the “Start Training” option from Case Editor actions:



Deep Learning Parameter Configuration

Max Epochs: The maximum number of iterations used by the identification.

Neurons: The number of neurons in each hidden layer.

Regularization: A weighting factor that minimizes the neural network coefficients.

Algorithm: The Deep Learning algorithm

GRU: Gated Recurrent Network

LSTM: Long Short-Term Memory Network

MLP: Multilayer Perceptron

Deep Learning Parameter Configuration (Cont..)

- **Parameter Count:** The number of internal model parameter that is required based on the current configuration. Clicking on the information icon will provide more detailed information.
- **Data Count:** The number of datapoints in all datasets. This number may be less than the actual number if oversampling has been requested.
- **Horizon (minutes):** The time to steady state of the deep learning model. The specified Horizon must be an integer multiple of the ID Period.
- **ID Period:** This is the sampling period of the datasets that should be used for identification.

Neural Network

More information is available about Neural networks below:

https://www.youtube.com/watch?v=oJNHXP50XDk&ab_channel=SteveBrunton

This video has variety of neural network architectures available to solve various problems in science and engineering.

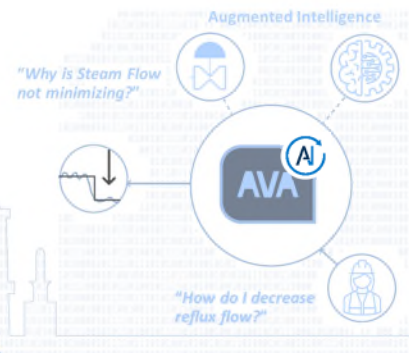


Aspen Virtual Advisor (AVA) for DMC3™

Elevate human-machine interactions with a **24X7 AI virtual advisor**

Find answers to common process control questions & get actionable guidance with AVA

**Achieve &
Maintain
Optimal
Operations**

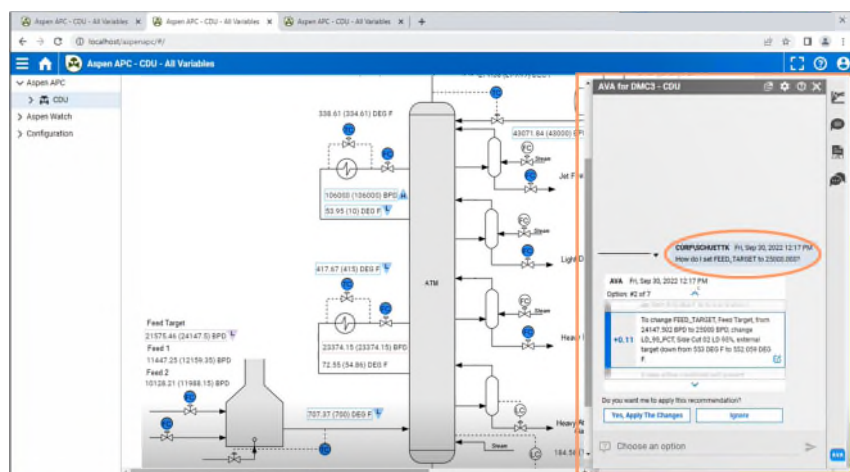


Enhance Operational Decision Making with Actionable Guidance & Real-time Augmented Intelligence

- Improve performance by elevating user understanding and interactions with DMC3

AVA provides guidance and ranked recommendations to meet operational targets

"How do I set CDU Feed Target to 25,000 BPD?"



Train New Engineers & Operators on DMC3

Aspen Virtual Advisor (AVA) for DMC3™

- Enable operators & engineers perform at a higher level

- **Capabilities:**

- Easy to use, on demand, chatbot-like UI
- Prompts help user build a question
 - Ask **“HOW DO I”** to get actionable recommendations to achieve optimal operating conditions



1

AVA for DMC3 - COLS3LOCAL

AVA: Sun, Jul 17, 2022 11:51 PM
Let me help you build your question!

How do I... Why is...
INCREASE DECREASE SET VALUE

FIC-2001 - FIC-2001

2.5

How do I set FIC-2001 to 2.500?

2

AVA for DMC3 - COLS3LOCAL

COMPOUNDETTE: Sun, Jul 17, 2022 9:52 PM
How do I set FIC-2001 to 2.500?

To change FIC-2001 from 2.628 MBBL/D to 2.5 MBBL/D, change AI-2022 high limit up from 4.5 MOLE % to 4.576 MOLE %

+1.0

Do you want me to apply that recommendation?

Yes, Apply The Changes Ignore

Choose an option

3

To change FIC-2001 from 2.628 MBBL/D to 2.5 MBBL/D, change AI-2022 high limit up from 4.5 MOLE % to 4.576 MOLE %

+1.0

To change FIC-2001 from 2.628 MBBL/D to 2.5 MBBL/D, change AI-2020 high limit up from 3.5 MOLE % to 3.549 MOLE %

+0.61

To change FIC-2001 from 2.628 MBBL/D to 2.5 MBBL/D, change FIC-2004 low limit up from 5 MBBL/D to 5.136 MBBL/D

-0.05

To change FIC-2001 from 2.628 MBBL/D to 2.5 MBBL/D, change AI-2021 low limit up from 2.5 MOLE % to 2.814 MOLE %

-0.18

Aspen Virtual Advisor: Quick Demo

Lesson Summary

In this lesson, you have learned about:

- The role of AI/ML in advanced process control
- Aspen Maestro Model
- Aspen Deep Learning for DMC3 and AspenIQ